

Evaluating Pre-trained Visual Representations with Diffusion Policies

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Abstract: This study investigates the integration of pre-trained visual representations (PVRs) into diffusion-based reinforcement-learning visuomotor policies to enhance robotic control by improving generalization and potentially reducing training requirements. Diffusion policies for robotic reinforcement learning, which generate robot actions through denoising processes, traditionally rely on end-to-end trained visual encoders. In recent years, diffusion policies have shown great success in many tasks, but have required significant computational and data resources to train. By embedding PVRs such as DINOv2, R3M, and ImageNet into these policies, we aim to lower the required training resources needed for these models and also improve the generality of these diffusion policies by allowing them to use more general representations. In our work, we evaluate their performance in the simulated Push-T environment. Our results indicate that frozen PVRs consistently underperform compared to end-to-end trained encoders. Our results indicate the limitations of PVRs when used without fine-tuning on specific datasets.

Keywords: Diffusion, Pre-trained Visual Representations, Robot Manipulation

1 Introduction

Developing general robots that can robustly operate in unpredictable environments and perform various tasks, remains an ongoing challenge in robotics. General sensorimotor policies require action from perception without using hand-tuned and highly engineered solutions for a broad range of environments and tasks. However, the challenge of training these policies for general sensorimotor control is often restricted by the need for extensive task-specific data, hindering performance in out-of-distribution (OOD) settings and thereby limiting the overall generality of the models.

Recently, there has been great interest in using the idea of foundational models, which leverage large internet-scale data to develop models that can successfully perform multiple downstream tasks in the fewshot or even zeroshot fashion, for tackling the problem of generalization in robotics [1]. Such models beckon the need for robust latent representation that accurately captures relevant features across many domains. Building on this idea, in this work we strive to analyze Pretrained Visual Representations for diffusion policies to assess how these impact task performance.

Pre-trained visual representations (PVRs) [2] have emerged as a solution in the vision community, offering general-purpose features that capture rich visual information from large datasets to enable model transferability across tasks. Foundational reinforcement learning models for robot action can

be used for a variety of tasks and with a variety of modeling techniques, but for a single policy to be capable of general action, it requires general representations as input. As a way to react to the need for diverse inputs, diffusion-based policies have shown success in solving a variety of robot action problems with simple visual inputs using visuomotor policies. Thus, we decide to investigate combining the positive attributes of all 3 methods to yield better success.

We propose that by embedding PVRs in visuomotor policies, robots can potentially achieve more robust and adaptive behavior without requiring task-specific visual encoders to be trained from scratch during the policy training process. This motivates our work to explore whether integrating PVRs into diffusion-based visuomotor policies can improve performance, improve generalization to novel environments, and reduce training time. Through systematic evaluation, our objective is to discover how different PVRs impact the effectiveness of visuomotor policies. In this work, we specifically investigate the improvement of previous work using diffusion-based visuomotor policies [3]. In our work, we test with four PVRs: DINOv2 [4] (in both large and base flavours) & ResNet-18 (using R3M [5] weights and IMAGENET1K_V1 [6] weights).

1. How does the use of PVRs compare to training an end-to-end encoder in diffusion policies?
2. How do different PVRs in diffusion policies compare to each other in performance?

2 Related Works

The design of PVRs [2] for sensorimotor control has been a promising development toward general-purpose visual perception for robotics. Their introduction helps to overcome the problem of limited data for specific tasks and addresses the ability of robot policies to generalize to new scenes. These PVRs are made by training neural networks that embed camera images into a visual feature space on huge amounts of data. These visual feature spaces can then support policy learning for a variety of robotic tasks such as locomotion, navigation, and manipulation. PVRs are designed to be general-purpose vision backbones for a diverse set of robotic tasks.

Prior work has shown that using PVRs helps in classical robot learning settings such as imitation learning [7] and reinforcement learning [8] can produce equivalent, or greater performance than end-to-end training of task-specific visual encoders.

In our work, we concretely perform experiments on the Diffusion Visuomotor Policy [3], which introduces a new form of robot visuomotor policy that generates behavior via a conditional denoising diffusion process on the robot action space. In this formulation, instead of directly outputting an action, the policy infers the action-score gradient, conditioned on visual observations, for K -denoising iterations. The visual encoder in this work, a standard Resnet-18 is trained end-to-end along with diffusion policy. They also report preliminary ablation studies, where they evaluate the success of two PVRs by training for smaller amounts of epochs; finding that across-the-board training with the frozen PVR encoder yields poor performance. Our work aims to replicate and extend these findings by adding more powerful PVRs such as the DinoV2 family, as well as training for longer epochs, for a fairer comparison against the end-to-end baseline, which was trained for 3000 epochs. We aim to generate further results to better understand why diffusion policy seems to diverge from other reinforcement-learning policies when it comes to leveraging pre-trained-visual encoders.

3 Method

3.1 Diffusion Model

Our testing framework utilized in our work builds on the diffusion visuomotor policy developed by Chi et al. [3]. The network for visuomotor robot policy learning was implemented using Denoising Diffusion Probabilistic Models (DDPMs), adapted to predict robot actions conditioned on visual observations. DDPM iteratively denoises Gaussian noise through a noise prediction network, ϵ_θ , to generate action sequences. Although both CNN and Transformer architectures for ϵ_θ net-

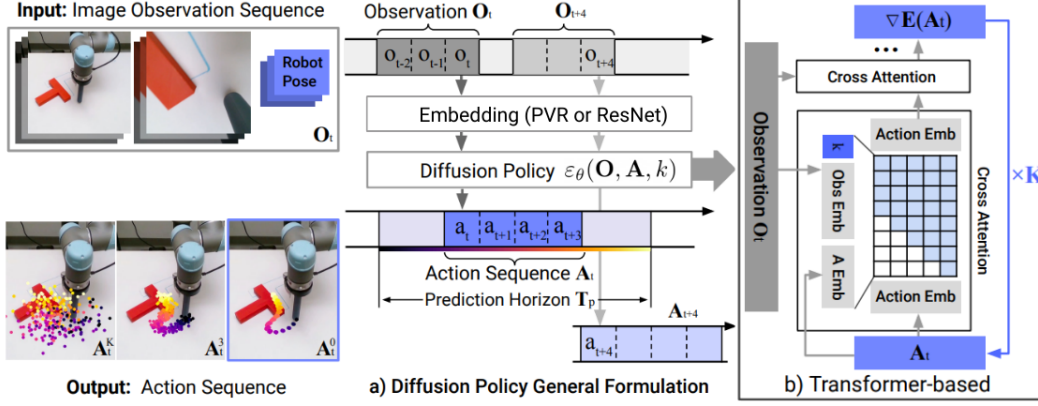


Figure 1: Diffusion Policy Overview from [3] modified to our policy.

works were explored in the work, it was found that the Transformer model performed better. As such, in our work we only use the transformer network in our studies. The transformer network designed by Chi et al. was trained to minimize the mean squared error (MSE) between the noise added to the input data and its prediction of that noise. For each training sample, a random denoising iteration k was selected and Gaussian noise ϵ_k with variance corresponding to iteration k was added to the original data. The transformer network ϵ_θ then predicted the added noise, conditioned on the observation sequence O_t and the denoising step k . The loss function is defined as: $L = \text{MSE}(\epsilon_k, \epsilon_\theta(O_t, A_t^0 + \epsilon_k, k))$, where A_t^0 represents the initial noisy action sequence. This loss ensures that the network learns to accurately estimate the noise at each step of the diffusion process. By minimizing this loss, the model approximates the conditional distribution $p(A_t|O_t)$, allowing it to generate temporally consistent action sequences conditioned on visual observations for real-time control of robotic agents. Additional details of the implementation of the DDPM policy can be found in the original document.

The DDPM policy was originally designed to use a ResNet-18 encoder to process visual input from raw input images. Our objective was to evaluate the performance of their diffusion policy in combination with various PVRs. To perform our analysis, we replaced the ResNet-18 encoder with encoding layers derived from the selected PVRs for each camera frame. These PVRs were frozen during training to assess whether the existing encodings of the PVRs were capable of performing well in the framework without adjusting them similar to how PVRs had been used in previous work in traditional robotics learning [7, 8].

3.2 Pre-trained Visual Representations

In this section, we briefly describe the chosen PVRs for this study. Note that for R3M and ImageNet we are referring to a ResNet backbone that has been trained on the corresponding datasets, however for brevity we will refer to them only as R3M & ImageNet.

3.2.1 R3M

Introduced in [5], the Reusable Representations for Robotic Manipulation is a unique PVR as it specifically targets the realm of robotics. In their work, they argue that to learn features useful for the manipulation setting, instead of using a general dataset like ImageNet they use the Ego4D dataset [9] which contains ego-centric clips of people performing miscellaneous tasks paired with a text description, totaling 3500 hours of data. Additionally, R3M designs a time contrastive learning cost function that is coupled with regularization to reduce the effective dimensionality of the learned embeddings. R3M supports multiple vision encoders as a backbone, in this case, we choose the ResNet-18 for direct comparison with the diffusion policy baseline.

3.2.2 ImageNet

The ImageNet dataset is a standard dataset [6] in computer vision. It organizes images by concepts in a hierarchical manner, it contains 21,841 concepts, each with an average of 650 images, totaling 14,197,122 images. The images were sourced from internet search engines and validated using Amazon Mechanical Turk.

3.2.3 DINOv2

DINOv2 [4] model uses the core idea of applying self-supervised learning for a vision transformer, which in practice is a BERT-like encoder, to enable learning on a scale from unstructured and unlabeled data, obtained by curating repositories of public web-crawled data. In this work, we test with two variants of the DINOv2 model, a base variant with 86.6 million parameters and a large variant with 304 million parameters.

4 Experiments

4.1 Environments

Our experiments are carried out in the Push-T environment. PushT, which is introduced in [3], involves pushing a T-shaped block to a fixed target using a circular end effector. Each run begins with randomized initial conditions for both the T-shaped block and the end effector. This task requires leveraging complex, contact-rich object dynamics to accurately push to the target position. The model receives RGB image observations augmented with proprioceptive data indicating the end-effector’s location.

5 Evaluation Setup

As a baseline for PushT, we train the randomly initialized ResNet-18 visual encoder in an end-to-end fashion following the training recipe from the original paper [3], training for a total of 3000 epochs. For each PVR, we replace the visual encoder for the chosen representation and freeze it during training. We repeat each modality across 3 seeds, except the DINOv2 family, in which we train only two seeds due to computational and time constraints. As the evaluation metric, we average the max return of the policy across the 10-latest epochs, averaging this value across all seeds. We also analyze 3 rollouts per seed and take the average duration to complete the task.

We note that we specifically do not finetune as we want to assess the off-the-shelf performance of the PVRs; given that this would represent the simplest and most generalizable use case for this type of representations.

5.1 Implementation Details

We train all models using a single 4090-RTX graphics card. Hyperparameters are taken from the original diffusion policy paper and are kept the same.

6 Evaluation on PushT

Table 1 shows quantitative results for all PVRs. We can see the modality that achieves the best performance is the one that trains the visual encoder in an end-to-end fashion. We also note that interestingly R3M, which specifically targets robotic applications, shows less difference with respect to the end-to-end baseline; showing only a -15.445% percentual difference. In comparison, models trained with large-scale general data show higher performance degradation at -40.126% for ImageNet; -35.327% for DINOv2-base; and -31.497% for DINOv2-large.

Visual Encoder	Average Score $\uparrow$	Average Duration (seconds) $\downarrow$
End to End	0.851	23.95
R3M	0.720	30.00
ImageNet	0.510	30.00
DINOv2-base	0.551	28.875
DINOv2-large	0.584	28.25

Table 1: Push-T Simulated average testing score of each observation embedding for the DDPM.

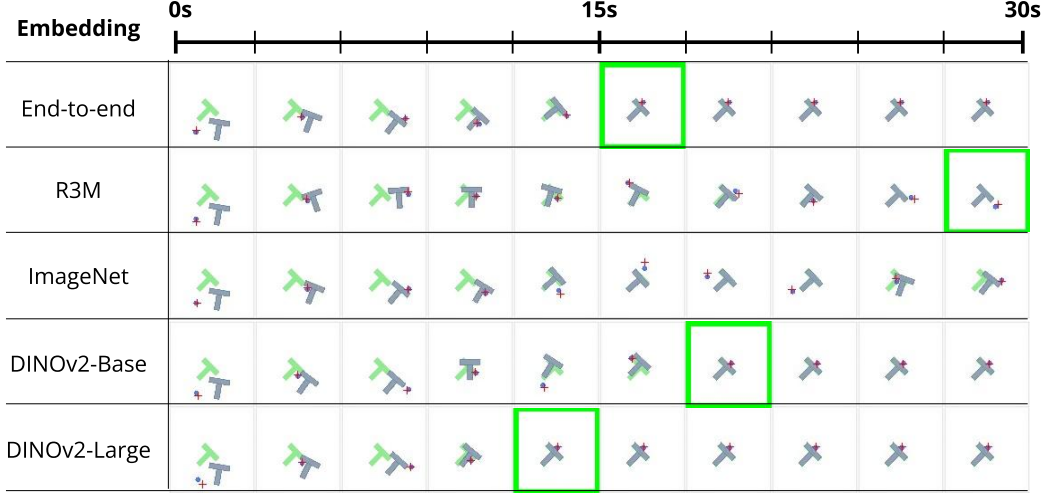


Figure 2: Consecutive observation frames for the best seeds of 4 PVR encodings and the baseline end-to-end model in the Push-T simulated experiment. The green box signifies when the model has reached the successful 95% area completion criteria. All remaining frames after success are repeated for consistency.

Figure 2 shows a roll-out of the diffusion policy using all 4 PVR encodings. We note that, empirically, we observe that on average models trained with an end-to-end visual encoder can complete the task in fewer timesteps than models using PVRs.

Figure 3 shows the mean score over the training epochs. Here we note that surprisingly, even in the early stages of the training there is no clear advantage of using PVRs. Thus, our initial hypothesis, that by leveraging pretrained knowledge in the visual encoder, we could achieve better performance with less training time, does not seem to hold. A possible explanation for this is that, given most of the tested PVRs are trained with general datasets, the fail to extract meaningful features for the task and simulation environment at hand.

7 Discussion

From these results, we now return to our initial questions in Section 1 to discuss our findings with intuitions for our results.

7.1 PVRs compared to End-to-end

Our experiments show a clear trend that using frozen PVRs in place of end-to-end learned encodings results in lower performance for diffusion policies. This is clearly seen in Figure 3 and Table 1, where we can see that the highest performance is seen in the end-to-end trained models as used in the original DDPM model from Chi et al. [3]. These results match the findings of Chi et al. in their ablation studies.

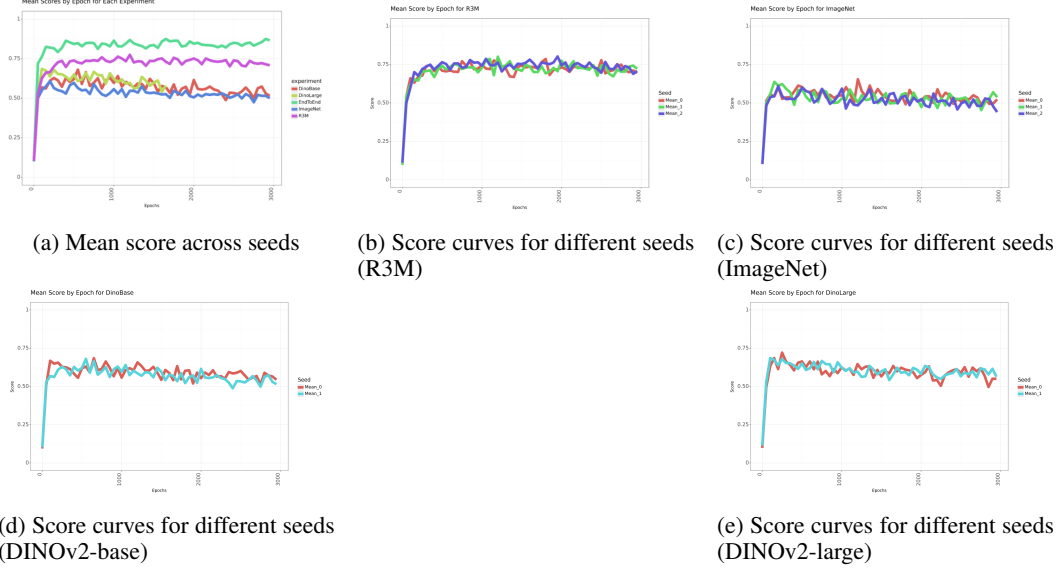


Figure 3: Score vs Epochs for PushT evaluation

This shows that the encoding used in the tested PVRs is significantly different from those preferred by the diffusion model we used for our Push-T trials. We believe that it is most probable that the difference in the method of generating the embeddings comes primarily from the differences in visual training data between the PVRs and the Push-T dataset, and not from an inherent property of diffusion policies. We believe this since we know that none of the training data for the PVRs included any tasks with high degrees of similarity to the Push-T task. Most of the PVRs were trained on data that was highly dissimilar to this task, and we believe that that is a primary reason for the decreased performance. In future work, we would like to explore fine-tuning PVRs with task-specific data to analyze this further.

7.2 PVRs compared to one another

Despite obvious performance gaps between PVR-embedded models and end-to-end DDPM, we see that certain PVR models demonstrate superior performance compared to others. In particular, as seen in Figure 1, R3M significantly outperforms both the DINOv2 models and ImageNet. There are multiple possible reasons why this could be, but we believe it is predominantly because the data used for R3M are more similar to the data found in the Push-T simulation environment. In all of our trials, we used simulated Push-T environments with static cameras. DINOv2, and ImageNet both use only real-world camera data as they are just trained on large-scale internet data. R3M, however, focused entirely on egocentric manipulation data which is relatively similar to our task, while the other PVRs include many other tasks outside of just manipulation and include much less egocentric manipulation data. This leads to an important understanding for our work: when using diffusion models, the PVR representations do not do well with out-of-distribution camera images. This means that PVR representations are not general enough for the diffusion models to use effectively unless part of the PVR training contains data on the task being performed.

8 Future Work

Our observations have important implications for future work with PVRs in diffusion models. We suspect for our conclusions that PVRs could reach equivalent, or better performance, end-to-end training if the PVR model were fine-tuned on task-specific data for a few hundred iterations before being used. Importantly, this is different from unfreezing the weights of the PVRs during training of our diffusion policy. We believe that frozen PVRs are still capable of effective use in diffusion

policies if independently fine-tuned on a small amount of task-specific data beforehand. Due to the high level of computational resources required for these trials alone, we were unable to explore this idea in our work, but hope to explore this in future work.

9 Conclusion

This work investigates whether the integration of PVRs into diffusion-based visuomotor policies can improve performance by improving generalization. Motivated by the emergence of diffusion-based policies in real-world robot applications coupled with the generalizing capabilities of foundation models, we sought to combine these features to yield stronger performance for state-of-the-art diffusion-based policies. Our results show that for the chosen diffusion policy, all PVRs clearly underperform the end-to-end trained diffusion model.

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